

A Multimodal Dynamic Hand Gesture Recognition Based on Radar–Vision Fusion

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[Haoming Liu](#) ; [Zhenyu Liu](#) [All Authors](#)

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Abstract:

Regarding increasingly complex scenarios in hand gesture recognition (HGR), it is challenging to implement a reliable HGR due to the nonadaptability of individual sensors to the environment and the discrepancy of personal habits. Multisensor fusion has been deemed an effective way to overcome the limitations of a single sensor. However, there is a lack of research on HGR to effectively establish bridges linking multimodal heterogeneous information. To address this issue, we propose a novel multimodal dynamic HGR method based on a two-branch fusion deformable network with Gram matching. First, a time-synchronized method is designed to preprocess the multimodal data. Second, a two-branch network is proposed to implement gesture classification based on radar–vision fusion. The input convolution is replaced by the deformable convolution to improve the generalization of gesture motion modeling. The long short-term memory (LSTM) unit is used to extract the temporal features of dynamic hand gestures. Third, Gram matching is presented as a loss function to mine high-dimensional heterogeneous information and maintain the integrity of radar–vision fusion. The experimental results indicate that the proposed method effectively improves the adaptability of the classifier to complex environments and exhibits satisfactory robustness to multiple subjects. Furthermore, ablation analysis shows that deformable convolution and Gram loss not only provide reliable gesture recognition but also enhance the generalization ability of the proposed methods in different field-of-view scenarios.

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SECTION I.
Introduction

With the rapid development of communication technologies [1] and the Internet of Things (IoT) [2], many sensing devices have been deployed to establish omnidirectional 3-D human–computer interaction (HCI). Hand gesture recognition (HGR), which is a noncontact HCI method with a high frequency of use and a strong expressive ability, has become a popular research topic. HGR has been beneficial in facilitating human daily life and has important applications in the interactive control of relevant scenes, such as vehicle-assisted driving [3], intelligent homes [4], sign language interaction [5], intelligent robot control [6], virtual reality [7], and medical surgery assistance [8]. Moreover, noncontact measurement-based HGR is very significant for avoiding the spread of global public health emergencies such as monkeypox and COVID-19 [9].

Gesture measurements can be divided into two main categories: contact devices and noncontact devices. Contact devices require equipment worn on the body to collect hand motion data, such as surface electromyography (SEMG) signals [10], acceleration, and motion direction [11]. However, the equipment is inflexible to wear and exposes users to the risk of bacteria or viruses. In contrast, noncontact devices, such as Wi-Fi-based, vision-based, and radar-based sensors, can efficiently capture a variety of gestures and prevent physical interaction with the machine surface. However, due to the limitations of Wi-Fi parameters [12] as well as electromagnetic interference and multipath reflection in the process of propagation, it is difficult to distinguish fine-grained gestures through Wi-Fi-based methods.

As a common device for gesture acquisition, vision-based sensors are roughly composed of RGB cameras and depth cameras [13], [14]. For instance, vision data gathered by the Kinect device are used to develop a natural gesture user interface for real-time tracking and gesture recognition [15], which relies on directed search of the entire hand outline to recognize gestures. Compared with the high computational complexity of the depth camera, the color camera can capture hand imaging at a lower power consumption [16]. In addition, due to the advantages of high frequency, short wavelength, and broad bandwidth [17], millimeter-wave (MMW) radar has satisfactory spatial resolution to distinguish gestures [18]. Google's Soli project [19] produces a frequency-modulated continuous wave (FMCW) radar sensor to detect gestures. In recent years, radar-driven gesture recognition has been used in vehicle entertainment systems and smart homes [20].

In recent decades, several algorithms have been used to model HGR tasks, such as Bayesian networks and HMMs [21]. These methods are only suitable for simple gesture classification. Regarding complex HGR tasks, high-dimensional features are roughly extracted by convolutional networks (CNNs). For instance, RGB images and depth images have been jointly inputted into a gesture classification model composed of a 3-D-CNN and a convolutional long and short-term memory (LSTM) network to achieve isolated gesture recognition [22]. A lightweight multichannel CNN has been presented to learn and classify radar-based gesture features, which are composed of radial distance, Doppler, and tangential azimuth–elevation [23]. In addition, the Gram matrix is proven to be an effective method that extracts structural information [24]. The Gram matrix matched by dense interleaving is also used to fit the feature gap between clean and degraded images [25]. Recently, Gram matrix matching has been used to maintain the complementary structure of remote sensing data [26].

However, regarding increasingly complex scenarios of HGR, such as light-limited nighttime, bright outdoor, car interiors, smart homes with multiradar working together, and indoor with high background noise, single vision-based or radar-based sensors do not reliably work in all these scenarios. On one hand, visual sensors can provide good imaging information at low costs with sufficient optical information [27], but vision-based gesture acquisition systems have also been greatly affected by lighting intensities [28]. On the other hand, energy scattering of gestures is captured by radar sensors to extract the radial motion information [29]. Regarding scenarios with high background noise, the distribution of gestures in radar data will be drowned out because gesture is a kind of micromotion with weak spectrum energy [30]. Therefore, complex scenarios will make a single sensor incapable of meeting the needs of reliable gesture acquisition.

HGR has also been confused by the variability and ambiguity of gesture measurements in different distance field of view (FOV) scenes. For cameras, near big and far small will lead to significant differences in the imaging size and details of the hand. For FMCW radar, different angles will lead to changes in the measurement direction and range of the radial distance or Doppler. The energy of the radar signal will attenuate with an increasing distance. Some studies have been proposed to address this issue. For instance, the RDAE domain is transformed to a 3-D Cartesian coordinate to improve the spatial consistency of multilocation datasets [31], and deformable convolution has been embedded in the separable 3-D-CNN (S3-D) to solve the spatiotemporal deformation caused by individual habits [32]. However, these solutions do not overcome the limitations of a single sensor.

Therefore, we propose a novel multimodal dynamic HGR method based on a two-branch fusion deformable network with Gram matching, which provides a reliable solution for increasingly complex HCI environments. First, a time-synchronized method is designed to preprocess the time and structure of multimodal data. Then, we propose a two-branch deformable HGR network with LSTM to address the gesture distortion in visual and radar data and further extract spatial–temporal features. Finally, Gram matrix matching is presented as a loss function to mine the heterogeneous information and fully exploit the complementary advantages of multimodal data. The main contributions of this article are as follows.

- 1) A novel dynamic HGR method based on radar–vision fusion is proposed to achieve deep linking of multimodal gesture information to cope with increasingly complex HGR environments. We achieve reliable gesture recognition in six kinds of environments consisting of variable lighting and different radar background noise.
- 2) Gram matrix matching driven by radar and vision data is first introduced into HGR classification to link high-dimensional heterogeneous information between multimodal data. To the best of our knowledge, this is the first work that calculates the Gram matrix to maintain the integrity of information fusion of multimodal HGR.
- 3) A two-branch deformable network with LSTM is proposed to enhance the weak generalization of HGR with different users and FOVs. This work completely takes advantage of the complementary heterogeneous information of different sensors while maintaining the temporal information of multimodal data.

The remainder of this article is organized as follows. Related works are introduced in [Section II](#). The fundamental and problem analyses are introduced in [Section III](#). The methodology is presented in [Section IV](#). The experimental setup and experimental discussion are shown in [Section V](#) and [Section VI](#), respectively. Finally, in [Section VII](#), concluding remarks are provided.

SECTION II.

Related Work

In this section, related works on vision-based HGR, radar-based HGR, and multimodal-based HGR are introduced.

A. Vision-Based HGR

The vision-based HGR can separate the hand from the environment through imaging information such as skin color, hand shape, and hand size [\[33\]](#). Early work adopted dynamic time warping (DTW) for gesture recognition [\[34\]](#). Then, hidden Markov models (HMMs), which mainly rely on the extraction of sign language direction, trajectory information, and synthetic shapes, were introduced for HGR [\[35\]](#). However, using HMM alone has limitations, especially in training context-sensitive models [\[36\]](#). In addition, some studies aim to estimate the state of hand joints to model 3-D gestures, such as fitting joint angles by hand contours [\[37\]](#) and extremity movement observation (XMOB) upper body trackers [\[38\]](#). Recently, Google’s media-tube project [\[39\]](#) calibrates the hand region at the acquisition end to help gesture modeling. Through the machine learning (ML) method of pretraining 30000 real and original images [\[16\]](#), the coordinates of 21 pixels of the hand in the image can be inferred to track the changes in the hand state in the real-time system.

With the emergence of inexpensive consumer depth cameras, gesture recognition systems using depth cameras have also been introduced [\[40\]](#). For instance, depth camera data are inputted into the lightweight convolution neural network (CNN) model to effectively detect and classify video-based gesture recognition [\[41\]](#). However, vision-based methods must address well-known environmental challenges, such as brightness, contrast, and exposure [\[42\]](#). Vision-based gesture acquisition systems have been greatly affected by illumination intensity. Regarding indoor or light-limited scenarios, reliable HGR is difficult to implement with a single visual sensor in a dark environment.

B. Radar-Based HGR

Recently, many radar-based methods have been studied to implement HGR through the range–Doppler–azimuth–elevation (RDAE) cube [43]. HGR based on a complementary multifeature fusion network (CMFF-HGR) [44] has been designed to extract RDA features of a range–time map (RTM), Doppler–time map (DTM), and angle–time map (ATM). These maps are obtained by fast Fourier transform (FFT) and multiple signal classification (MUSIC) algorithms [45]. However, this work only achieves classification in simple gestures. To cope with the classification of more gestures, spatiotemporal deformable and context-aware modules have been embedded into the classification model to mine the interrelation among time-varying 5-D feature spaces [32]. This work provides a standardized radar signal preprocessing procedure to encode multichannel feature information into a time-varying 5-D feature spaces. By introducing CumSoft blocks on top of recurrent neural networks (RNN), a gesture triggering strategy is presented to determine the appropriate time for early gesture prediction [46]. This work uses range–Doppler map (RDM) video to achieve early HGR. The recognition rate can reach 88.85%, while the average time savings are 69.47%. The time–frequency map and spectrogram videos are inputted into a dual-stream fusion deformable residual network embedded with a convolutional block attention module (CBAM) to implement robust HGR, which achieves 97.5% classification accuracy [47].

However, regarding environments with high background noise, radar-based methods may not be reliably implemented. For instance, two vertically [48] and three relatively [49], [50] placed radars have been used for capturing complementary writing data, concurrently operated radars have been deployed to detect kitchen activities [51], and multiple radar systems have been designed for multiple indoor tasks and vital sign monitoring [52], [53]. Multiradar working at the same time will produce significant mutual interference [54], which will increase the electromagnetic clutter energy and decrease the performance of radar perceptions [55]. Gesture, which is a kind of micromotion, has relatively weak spectrum energy. Therefore, the energy distribution of gestures will be obscured by high environmental background noise.

C. Multimodal-Based HGR

Compared with using only vision or radar sensors, multimodal fusion is considered to be an effective way to overcome the limitations of a single sensor. For instance, gesture data captured by an ultrawideband (UWB) Doppler radar and wearable pressure sensors are inputted into an SVM classifier to achieve simple human microgesture recognition [56]. Data gloves composed of pressure and flexible sensors are designed to capture gestures, while these hand movements are classified by residual modules and LSTM [57]. These solutions inevitably face the limitations of wearable devices, while the combination of noncontact methods will not suffer from such problems. The multimodal cross-learning method uses radar RDM, thermal imaging camera images, and RGB camera images to train the automatic encoder model and only uses radar data for inference to achieve low-cost stove gesture control [58].

Some methods study the radar–vision fusion of autopilot in the roughly complex and changeable environment of roads. For instance, a method based on data-layer fusion is proposed: this method combines the weight and correlation consistency function of multisensor target recognition to construct the data fusion support computing model [59]. Because radar and vision sensors have different physical properties and regions of interest, data-layer fusion is difficult to directly implement. Another fusion method is decision-level fusion. Different networks are used to extract features from RGB images and radar spectrum maps for common target recognition [60].

However, in contrast to the recognition and segmentation of large rigid objects (such as cars, trees, and roads) in the autopilot scene, the general objects distinguished in the HGR scene are small nonrigid microgestures. Therefore, research on autopilot cannot be introduced for HCI. Time-of-flight (ToF) data have been proposed to be combined with radar data to achieve human tracking [61]. ToF provides information about the shape and exact position of the target in the 3-D world, which is confirmed by radar and provides additional information about the speed of each moving target. However, this work is not applied to microgesture recognition. In addition, although the NVIDIA institute presents a data-layer fusion HGR method by directly inputting radar data, RGB camera data, and depth camera data into the DCNN [62], they ignore that the characteristics of measurement data between sensors, such as the data size, data structure, and unrelated physical attributes, are always different.

The existing research focuses on HGR based on data fusion of wearable devices, radar–vision fusion for autopilot, and multimodal fusion for behavior tracking, while research on gesture recognition based on radar–vision fusion is still in its early stages. Moreover, existing studies do not provide sufficient attention to the information fusion of multimodal gesture data. In view of the increasingly complex noncontact HGR scenarios, it is important to excavate this high-dimensional structural information of multimodal data. In particular, regarding the HGR based on radar–vision fusion, a method that effectively establishes bridges linking heterogeneous attributes and complementary information between data of different sensors has not been investigated.

SECTION III.

Fundamental and Problem Analysis

Fundamental and problem analyses are introduced in this section.

A. Vision Model

According to the camera imaging principle and the similarity principle, the corresponding relationship between the image calibration $P(u, v)$ and the camera coordinate system $P(X_c, Y_c, Z_c)$ can be expressed as [63]

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{dx} & 0 & u_o \\ 0 & \frac{1}{dy} & v_o \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} \quad (1)$$

[View Source](#) 

where (u, v) , (x, y) , and (X_c, Y_c, Z_c) denote the pixel coordinate, the image coordinate, and the camera coordinate of point p , respectively; (u_o, v_o) denotes the origin of the pixel coordinate system; and f denotes the focal length of the camera. The obtained hand landmarks contain coarse motion information that are used to estimate the bounding boxes for hands, which cannot achieve the resolution of distinguishing fine-grained gestures. This is because the camera is adept in capturing the gesture azimuth, elevation, and morphological information, but lacks the ability to capture micromotion information.

B. Radar Model

FMCW radar has been used to capture gesture information such as range, velocity, and angle [23]. By mixing the received signal with a copy of the transmitted signal at the in-phase receiver and passing through the resulting signal to a low-pass filter, the decoupled intermediate frequency (IF) signal after sampling can be approximately expressed as [43]

$$b^{(i_f)}(i_s, i_c) = A \exp(j2\pi(f_r i_s P_s)) \cdot \exp(j2\pi(f_d i_c P_c)) \quad (2)$$

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where A , f_r , f_d , P_s , and P_c denote the amplitude of IF signal, the range frequency, the Doppler frequency, the sampling period, and the chirp period, respectively, and $i_s = 1, \dots, I_s$, $i_c = 1, \dots, I_c$, and $i_f = 1, \dots, I_f$ denote the index of the sampling points, the index of the chirps, and the index of the frames, respectively.

Through the FFT of different dimensions for the above signal model, the range and Doppler spectrum distribution of the frame-based radar signal can be expressed as

$$\begin{aligned} B_R^{(i_f)}(r, i_c) &= \sum_{i_s=1}^{I_s} b^{(i_f)}(i_s, i_c) \cdot \exp\left(-j2\pi \frac{r i_s}{I_s}\right) \\ B_{RD}^{(i_f)}(r, v) &= \sum_{i_c=1}^{I_c} B_R^{(i_f)}(r, i_c) \cdot \exp\left(-j2\pi \frac{v i_c}{I_c}\right) \end{aligned} \quad (3)$$

[View Source](#) 

where B_R denotes the range spectrum distribution. Moreover, B_{RD} denotes the range–Doppler spectrum distribution, which can be mapped by the energy to construct the RDM. $r = 1, \dots, R$ and $v = 1, \dots, V$ denote the range frequency index and the Doppler frequency index, respectively. The angle distribution can be estimated by the antenna dimensional FFT or direction of arrival (DOA) positioning methods, such as the MUSIC algorithm and Capon beam-forming algorithm.

C. Problem Analysis

Related work on vision- and radar- based HGR indicates that although cameras and radar have been used to implement hand gesture capture, cameras are good at collecting position and morphological information, while radar is good at collecting radial range and velocity information. The two sensors are highly complementary. On one hand, radar can provide radial information with higher resolution than depth cameras. On the other hand, the RGB camera can supplement the imaging information and obtain the position at a low cost. Therefore, it is sensible to combine cameras and radar for gesture acquisition.

The main challenge of HGR is to reliably implement gesture capture and recognition in complex environments. However, it is difficult for RGB cameras to work in brightly lit or dark environments. In crowded indoor or multiradar deployment scenes, the radar spectrum energy of gestures will be obscured by environmental clutter, which leads to the loss of the hand object. Therefore, the existing HGR studies based on a single sensor have difficulty achieving robust HGR in the above complex environments.

Another challenge of HGR is the ambiguity and variability of gestures performed by different people or in different FOV scenes. The habits and hand size of different humans will cause evident discrepancies in the range and speed of the same gesture. The different relative positions with the measurement platform will also lead to confusion between the gesture categories. The gesture classification errors caused by these distortions restrict the generalization performance of HGR.

SECTION IV. Methodology

Our work achieves dynamic gesture recognition that adapts to complex environment and based on two aspects. First, according to the displacement and energy differences of each frame of gesture, a preprocessing method is designed to synchronize the time and structure of vision and radar data. Second, we propose a novel multimodal dynamic gesture recognition method based on a two-branch fusion deformable network with Gram matching. The input layer of the two branches is replaced by the deformable layer. The spatial features of each moment are fed into the long short-term memory (LSTM) layer. Gram matrix matching is presented as a loss function to fully exploit the complementary advantages of multimodal data. An overview of the proposed method is shown in [Fig. 1](#).

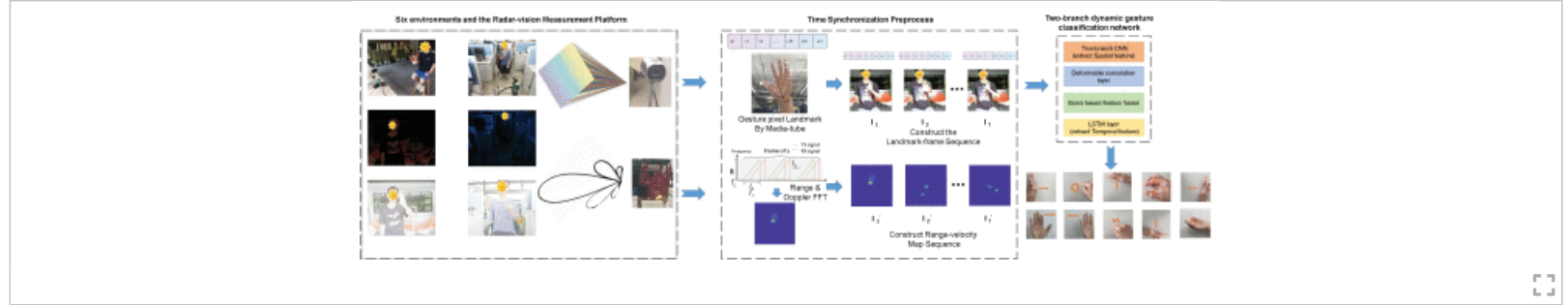


Fig. 1.
Overview of our proposed camera and radar-fusion-based dynamic gesture recognition method.

A. Time-Synchronized Preprocessing Method

In the application of HGR, dynamic gestures generally consist of the following three stages: the preparation stage, core stage, and retraction stage. The preparation and retraction stages of different gestures are roughly similar. The core stage contains the main information of gesture motion, such as different speeds and ranges of movement. Therefore, the extraction of the start and end time-stamps of the core stage is very important for gesture classification.

However, regarding multisensor data streams, the segmentation of gesture core stage is a difficult task because of the great differences in data structure and physical characters. Optical-based RGB video data reflect the morphological changes in the hand, while radar-based RDM data contain the difference in signal energy amplitudes of gestures.

Therefore, we propose a time-synchronized preprocessing method based on the RGB morphology and radar spectrum differences. To obtain the gesture state of each frame of visual data, 21 pixel coordinates of the hand are calculated at the end of acquisition. The coordinates can be expressed as $(x_i^{(k)}, y_i^{(k)}), i \in (1, 21), k \in (1, K)$, where i represents the i th coordinate, and k represents the k th frame. The change amplitude of the 21 coordinates with time can be regarded as the time index of the visual data. The displacement of these points between the k th and $(k + 1)$ th frames can be expressed as

$$d_i^{(k)} = \left\| \left(x_i^{(k)} - x_i^{(k+1)} \right) + \left(y_i^{(k)} - y_i^{(k+1)} \right) \right\|_2 \quad (4)$$

[View Source](#)

where $d_i^{(k)}$ represents the displacement of the i th coordinate in the k th frame. The displacements of the 21 coordinates with K frames are normalized

as

$$D_i^{(k)} = \frac{d_i^{(k)} - \min_{k \in K} (d_i^{(k)})}{\max_{k \in K} (d_i^{(k)}) - \min_{k \in K} (d_i^{(k)})} \quad (5)$$

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where $D_i^{(k)}$ represents the normalized displacement of the i th coordinate in the k th frame.

Then, the normalized displacement values of each coordinate of each frame are summed as

$$D_{\text{sum}}^{(k)} = \sum_{i=1}^N D_i^{(k)} \quad (6)$$

[View Source](#) 

where $D_{\text{sum}}^{(k)}$ represents the summation amplitude of the normalized displacement in the k th frame. When $D_{\text{sum}}^{(k)}$ is greater than the set threshold T_{vision} , the frame is marked as the start time-stamp $k_{\text{vision}}^{\text{start}}$ of the core stage. When $D_{\text{sum}}^{(k)}$ is less than T_{vision} , the frame is marked as the end time-stamp $k_{\text{vision}}^{\text{end}}$ of the core stage. After the above preprocessing, the RGB video data $X^V(C, K, H, W)$ with K frames and a size of $C \times H \times W$ have been converted into $X^V(C, k_{\text{vision}}^{\text{start}} : k_{\text{vision}}^{\text{end}}, H, W)$.

Similarly, the amplitude value of the radar spectrum is used to characterize the gesture state of each frame of radar data. First, the RDM of formula (6) is normalized as

$$S_{(u,v)}^{(k')} = \frac{s_{(u,v)}^{(k')} - \min_{k' \in K'} (s_{(u,v)}^{(k')})}{\max_{k' \in K'} (s_{(u,v)}^{(k')}) - \min_{k' \in K'} (s_{(u,v)}^{(k')})} \quad (7)$$

[View Source](#) 

where $S_{(u,v)}^{(k')}$ represents the normalized amplitude of the RDM in the k' th frame.

Then, the normalized amplitude values of the RDM of each frame are summed as

$$S_{\text{sum}}^{(k')} = \sum_{u=1}^U \sum_{v=1}^V S_{(u,v)}^{(k')} \quad (8)$$

where $S_{\text{sum}}^{(k')}$ represents the summation amplitude of the RDM in the k' th frame. When $S_{\text{sum}}^{(k')}$ is greater than the set threshold T_{radar} , the frame is marked as the start time-stamp $k_{\text{radar}}^{\text{start}}$ of the core stage. When $S_{\text{sum}}^{(k')}$ is less than T_{radar} , the frame is marked as the end time-stamp $k_{\text{radar}}^{\text{end}}$ of the core stage. After the above preprocessing is accomplished, the radar spectrum data $X^R(C, K', H', W')$ with K' frames and the size of $C \times H \times W$ have been converted into $X^R(C, k_{\text{radar}}^{\text{start}} : k_{\text{radar}}^{\text{end}}, H', W')$.

Finally, the size and the duration of different sensor data streams are synchronized. The multisensor real-time data streams are segmented to construct the core stage of gestures. The final start and end time-stamps are calculated by the minimum values in the gesture start and end time-stamps of different data streams as $k_{\text{start}} = \min(k_{\text{vision}}^{\text{start}}, k_{\text{radar}}^{\text{start}})$, $k_{\text{end}} = \min(k_{\text{vision}}^{\text{end}}, k_{\text{radar}}^{\text{end}})$. After the reconstruction of the same size of $c \times h \times w$, the synchronized visual and radar data are expressed as $(X^V(c, \Delta k, h, w), X^R(c, \Delta k, h, w))$, where $\Delta k = k_{\text{end}} - k_{\text{start}}$ represents the lasting time of the core stage.

B. Dynamic Deformable Gesture Recognition Network

Although time-synchronized preprocessing integrates the structure of two-stream data, deep fusion is restricted by the irrelevance of physical properties. Since RGB camera-based optical information and radar-based electromagnetic radiation information cannot be learned by a single network, a two-branch network structure is used to fully extract the multimodal spatial features of gestures. The two branches have the same structure, and the input convolutional layers are replaced by deformable convolutions to address the gesture distortion caused by different measurement conditions. Moreover, since a single CNN cannot exploit the temporal information embedded in the data, combinations of CNN and LSTM units have been used for temporal dynamic gesture modeling. Hence, we propose a two-branch deformable dynamic HGR network, as shown in Fig. 2.

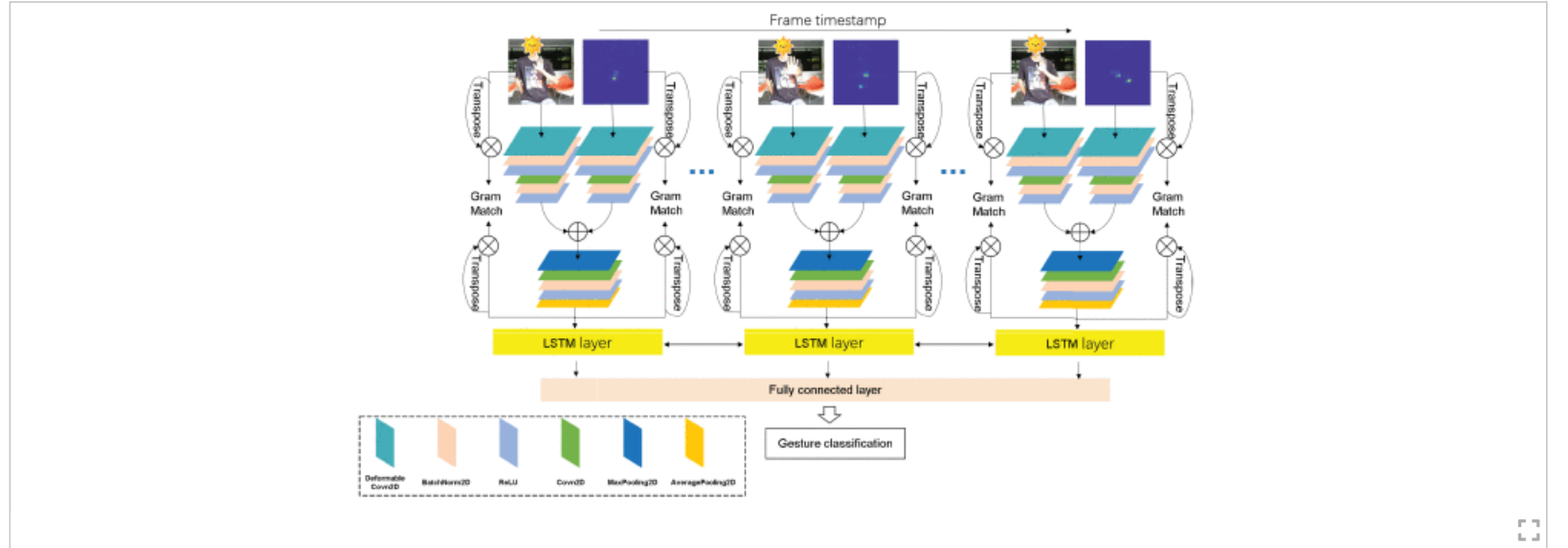


Fig. 2.

Proposed two-branch fusion deformable dynamic gesture recognition network based on Gram matching.

Specifically, the CNN consists of a 3×3 ReLu-activated convolution layer, a BatchNorm layer, and a 1×1 ReLu-activated convolution layer. The combination of the BatchNorm layer and the excitation function of ReLu accelerates the training speed of the depth neural network and has the characteristic of unilateral suppression. Then, the dual-flow features are cascaded and inputted into the fusion block to further extract the fusion feature map with heterogeneous features. The fusion block consists of a 3×3 max-pooling layer, a 1×1 ReLu-activated convolution layer, and a 3×3 average-pooling layer. The pooling operations have been applied in the fusion data to desample the feature size and improve the training speed.

Moreover, distinguishing gestures are roughly confused by different measurement conditions, such as different speeds, distances, and positions, especially in FOV scenes with various ranges. These distortions will lead to different sizes and positions of the HGR receptive field, semantic ambiguity, and other variability problems. To address these issues, the grid sampling of convolution is deformed freely via irregular offsets. We replace the input convolution layers with the deformable-2-D (D2-D) to further enhance the robustness of HGR, as shown in [Fig. 2](#). Kernel deformation is performed on the input data based on D2-D, which adds 2-D offsets to the conventional grid sampling position in standard convolution. Regarding the input feature X of size $C \times H \times W$, the reference channel dimension is assumed to be 1 because the channels of each input pixel share a set of offsets. Suppose a traditional convolution layer with n_k 2-D conventional kernels K of size $h_k \times w_k$. A 2-D convolution operation, which takes X as the input and outputs Y of each spatial position p_0 , is expressed as

$$\begin{aligned} Y_{2d}^{p_0} &= [y_{p_0}^1, y_{p_0}^2, \dots, y_{p_0}^{n_k}]^T \\ y_{p_0}^n &= \sum_{h,w} K_{h,w}^n \cdot X_{h,w}^{p_0} \end{aligned} \quad (9)$$

[View Source](#) 

where $y_{p_0}^n$ represents the output of the n th conventional kernel at p_0 . To release the limitation of a fixed square, the learnable offset is added to the convolution kernel to simulate complex geometric transformations, which can be expressed as

$$\begin{aligned} Y_{2d}'^{p_0} &= [y_{p_0}'^1, y_{p_0}'^2, \dots, y_{p_0}'^{n_k}]^T \\ y_{p_0}'^n &= \sum_{h,w} K_{h,w}^n \cdot X_{(h+\Delta H, w+\Delta W)}^{p_0} \end{aligned} \quad (10)$$

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where $\Delta H \times \Delta W$ denotes the size of the offset features. The learned offsets refine the input feature $X_{h,w}^{p_0}$ to $X_{(h+\Delta H, w+\Delta W)}^{p_0}$ to guide the deformation of the traditional grid sampling. By applying the D2D operation to RGB image and RDM spectrum data, the learnable offsets expand the receptive field and improve the generalization performance of gesture motion modeling.

Finally, after the fusion operation is performed, the features are fed into the LSTM layer to extract the temporal features of gesture motion. The LSTM is mainly composed of the following five parts: the input gate, the output gate, the memory cell, the forget gate, and the hidden state. The memory cell stores, modifies, and accesses the hidden state. An input sequence $x = [x_1, \dots, x_K]$ is assumed, where x_k denotes the feature vector extracted by the

CNN in the k th frame. Regarding each LSTM unit, the relationship between the input, hidden state, and output can be expressed as follows:

$$\begin{aligned}
I_k &= \sigma(X_k W_{XI} + H_{k-1} W_{HI} + C_{k-1} W_{CI} + B_I) \\
O_k &= \sigma(X_k W_{XO} + H_{k-1} W_{HO} + C_k W_{CO} + B_O) \\
C_k &= C_{k-1} F_k + I_k \tanh(X_k W_{XC} + H_{k-1} W_{HC} + B_C) \\
F_k &= \sigma(X_k W_{XF} + H_{k-1} W_{HF} + C_{k-1} W_{CF} + B_F) \\
H_k &= O_k \tanh(C_k)
\end{aligned} \tag{11}$$

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where σ denotes the logistic sigmoid function, and I , O , C , F , and H denote the input gate, the output gate, cell activation, the forget gate, and the hidden state, respectively.

C. Gram-Guided Loss Based on Multimodal Fusion

To fit the high-dimensional heterogeneous information between the input multimodal data and the fused feature maps, the Gram matrix is chosen as the loss function to further facilitate the maintenance of the complementary structure of fusion feature maps. In contrast to the dense interleaving method that matches the Gram matrix, the Gram matrices of the RGB camera source and the radar spectrum source are referred to as the texture structure and the spectrum structure of the fusion feature, respectively.

First, as shown in [Fig. 2](#), the neural activated Gram matrix is used to extract heterogeneous information from different sources and to maintain the complementary structure of the fusion information. Moreover, $V = [V_1, V_2, \dots, V_{n_d}] \in R^{O \times P}$ denotes an RGB camera dataset with O samples of n_d class, and $R = [R_1, R_2, \dots, R_{n_d}] \in R^{O \times P}$ denotes the FMCW radar dataset. Because of the synchronous preprocessing operation, the samples of the two datasets have the same dimension P , which is obtained by the width $\times$ height of the feature maps. It is supposed that $f(\cdot)$ is a compound function of consecutive operation in one hidden layer, $c^{(l)}$ and $s^{(l)}$ denote the number and size of the hidden features in the l th layer, respectively, and $F^{(l)}(V, R) = f^{(l)}(\dots(f^{(1)}(V, R)))$ denotes the hidden features achieved in the l th layer. Regarding RGB camera data, each element in the Gram matrix $\mathcal{G}$ can be defined as

$$\begin{aligned}
\mathcal{G}_{i,j}(V) &= \sum_{p=1}^P V_{i,p} V_{j,p} \\
\mathcal{G}_{i,j}(F^{(l)}(V, R)) &= \sum_{p=1}^P F_{i,p}^{(l)} F_{j,p}^{(l)}
\end{aligned} \tag{12}$$

[View Source](#) 

where i, j represent the i th and the j th points of the feature maps, respectively, and $\mathcal{G}_{i,j}(V)$ and $\mathcal{G}_{i,j}(F^{(l)}(V, R))$ indicate the value of the inner product operation of the feature maps of the corresponding points in the input and fuse layers, respectively. Because $\mathcal{G}_{i,j}(\cdot)$ jointly activates the high-dimensional information of the feature maps, the texture structure of the RGB camera data has been well obtained.

Second, the spectrum structure is captured from the radar spectrum map data using the Gram matrix $\mathcal{G}$ operation by the inner product of the corresponding feature maps. Each element in the Gram matrix $\mathcal{G}$ can be defined as

$$\begin{aligned}\mathcal{G}_{i,j}(R) &= \sum_{p=1}^P R_{i,p} R_{j,p} \\ \mathcal{G}_{i,j}(F^{(l)}(V, R)) &= \sum_{p=1}^P F_{i,p}^{(l)} F_{j,p}^{(l)}\end{aligned}\quad (13)$$

[View Source](#) 

where $\mathcal{G}_{i,j}(R)$ and $\mathcal{G}_{i,j}(F^{(l)}(V, R))$ indicate the structure relationship of the feature maps of the corresponding channels in the input and fuse layers, respectively.

Third, to measure the gaps between the input data from different sources and the feature map of the fusion layer, the texture and spectrum metrics are defined by loss terms as

$$\begin{aligned}\mathcal{L}_{\text{tex}} &= \frac{1}{4P^2O^2} \sum_{i=1}^P \sum_{j=1}^P \left(\mathcal{G}_{i,j}(V) - \mathcal{G}_{i,j}(F^{(l)}(V, R)) \right)^2 \\ \mathcal{L}_{\text{spe}} &= \frac{1}{4P^2O^2} \sum_{i=1}^P \sum_{j=1}^P \left(\mathcal{G}_{i,j}(R) - \mathcal{G}_{i,j}(F^{(l)}(V, R)) \right)^2\end{aligned}\quad (14)$$

[View Source](#) 

where $\mathcal{L}_{\text{tex}}$ and $\mathcal{L}_{\text{spe}}$ represent the RGB camera and radar spectrum guided texture and spectrum loss, respectively. Due to the synchronous preprocessing and the same dual-branch convolution operation, the dual-stream feature distributions with the same number of samples O and dimensions P are consistent.

Finally, regarding the gesture classification task, the cross-entropy operation is used to measure the gap between the predictive value y_c and the real value, which is defined by loss terms as $L_c = -\sum_{n=1}^{n_d} y_c \log(p_c)$, where p_c represents the predictive probability for the category, and M represents the number of categories in the classification task. Hence, the overall training loss of the proposed model is the combination of the weighted Gram-guided loss and the cross-entropy loss

$$L = \lambda(\mathcal{L}_{\text{tex}} + \mathcal{L}_{\text{spe}}) + L_c \quad (15)$$

[View Source](#) 

where λ is a balance parameter. Since the Gram operation can be deemed a form of regularization, λ should be adjusted to avoid overregularization in the process of dual-flow fusion.

[Fig. 2](#) shows the end-to-end framework of the proposed two-branch fusion deformable dynamic gesture recognition network based on Gram matching. Each frame of the RGB image and RDM spectrum belonging to the same gesture sequence are extracted and fused by a two-branch CNN. Then, the fusion output is fed into the LSTM units to extract the timing features. The LSTM output vectors of all the moments are aggregated through the fully

connected layer, and then a gesture label is predicted. The loss value of the predictive result is calculated, and then back propagation is used to calculate the gradient.

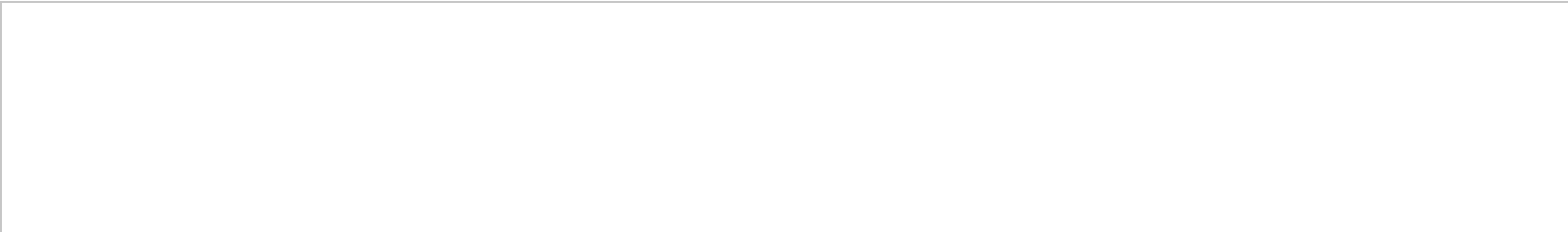
SECTION V.

Experimental Setup

To evaluate the complex environment adaptability and generalization of the proposed HGR method, we carried out a large number of experiments on the multisensor measurement platform composed of an RGB camera and FMCW radar, as shown in Fig. 3. The radar sensor used in the platform is the 77-GHz FMCW radar of Texas Instruments AWR1843BOOST, whose parameters are set as shown in Table I.

TABLE I Parameter of FMCW Radar Used in Measurement Platform

Radar Parameters	Value
Start Frequency (GHz)	77
Frequency Slope (MHz/ μ s)	30
Sample Rate (Ksps)	2048
Number of ADC Samples	256
Number of Chirp Loops	128
Bandwidth (GHz)	3.96
Range Resolution (cm)	4
Velocity Resolution (cm/s)	10
Chirp Cycle Time (μ s)	160
Frame Cycle Time (ms)	100



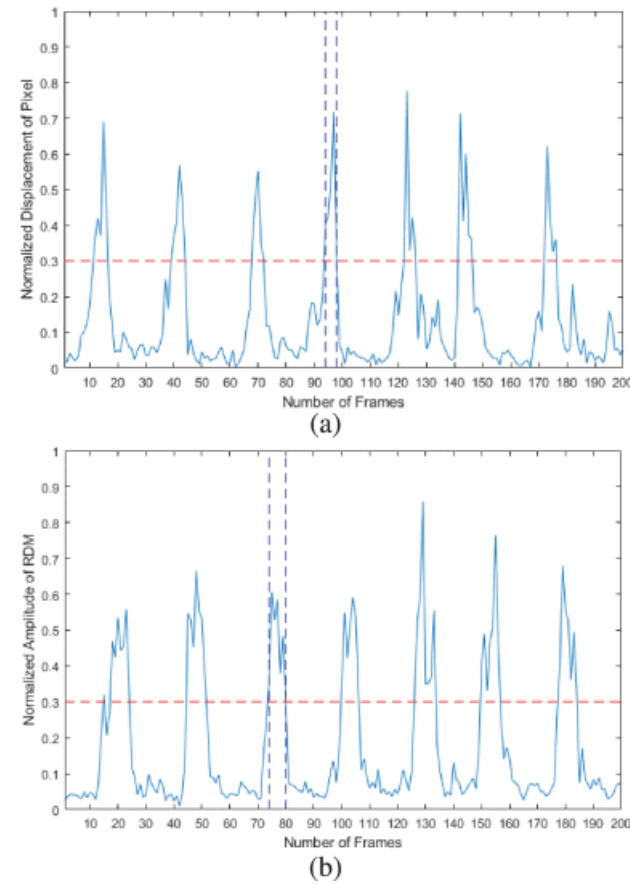
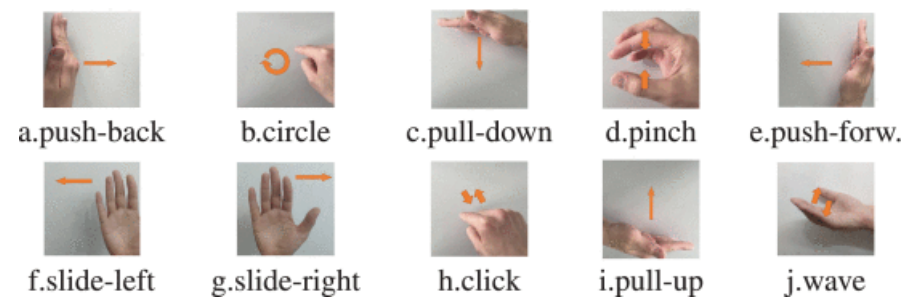


Fig. 4.

After the time-synchronized preprocessing, the lasting time of camera and radar data streams are synchronized. The red dotted lines denote the value of the set threshold, and the blue dotted lines denote the start and end time-stamps of the gesture. (a) Displacement of gesture pixels extracted from the each RGB frame. (b) Amplitude of gesture spectrum extracted from each RDM frame.

These gestures are captured by eight subjects to establish a train-validation dataset with 2400 samples. Each sample is composed of two-part contexts of image and RDM. Considering the individuals of different subjects, the eight times leave-one-out cross-validation method was used to evaluate the robustness of the proposed method. Taking into account the different environments and measurement scenes, hand data were collected by four extra subjects in six environments to build a test dataset with 7200 samples. Six kinds of environments, including out-of-laboratory scenes of normal light, dark and bright light, and in-laboratory scenes of normal light, dark and bright light, are deployed, as shown in Fig. 5. The relative position between each subject and the platform is determined according to the habits of the subjects, and it is greater than 1 m. In addition, a test dataset with 1200 samples are collected at four distance FOV scenes, i.e., 1, 1.5, 2, and 2.5 m, from the measurement platform in and out-of-laboratory scene of normal light, to verify the generalization performance of the proposed network.

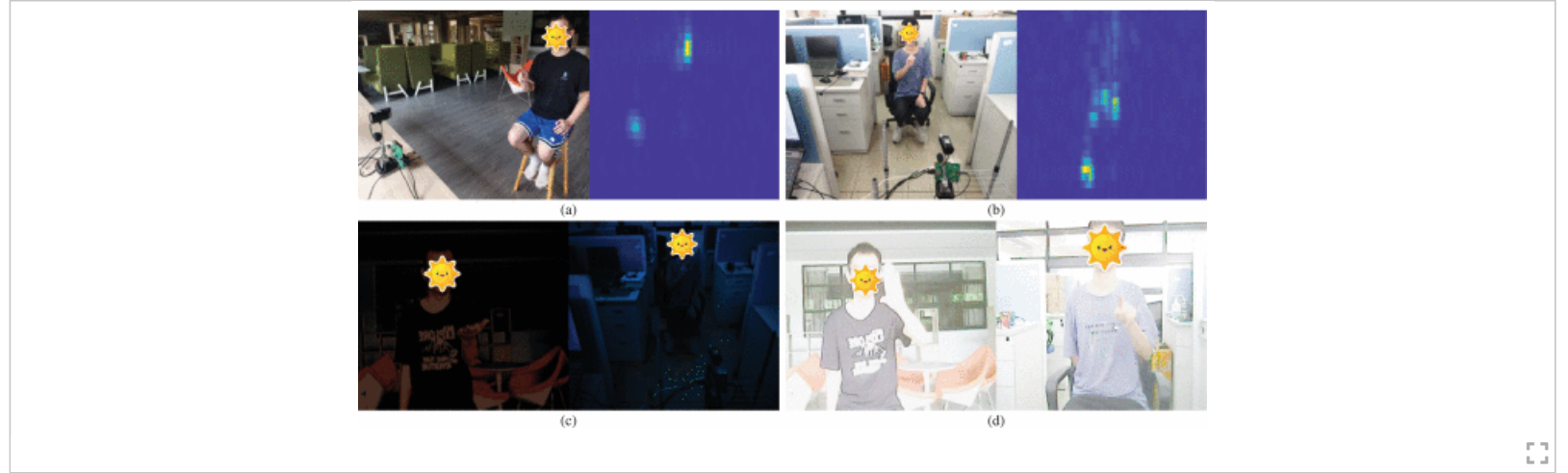


Fig. 5.

Six kinds of environments considered in our works. (a) Out-laboratory of normal light. (b) In-laboratory of normal light. (c) Out- and in-laboratory of dark. (d) Out- and in-laboratory of bright light.

SECTION VI.

Experiment Analysis and Discussion

In this section, the advantages of the proposed method in complex environment adaptability and generalization are verified. To evaluate the fitting degree of the training process, accuracy, average recall, average precision, and average F1-score are used as metrics, the initial learning rate is set to 0.1, and the balance factor is set to 0.01. The entire experiments are implemented in a computer with an Intel i7-8750H processor and an Nvidia Geforce GTX 1050Ti GPU card.

A. Leave-One-Out Cross-Validation

Due to the differences in the personal habits of the subjects, gesture samples of the same category may express different meanings. Therefore, satisfactory generalizability to different users is significant for the HGR. We use leave-one-out verification to evaluate the generalization performance of

the proposed method. Regarding each leave-one-out validation, one subject who did not participate in the training was selected as the validation set of the unknown source, and the remaining seven subjects were selected as the training set. The accuracy, average recall, average F1-score, and average precision of gesture classification for each subject are provided to evaluate the generalization. Meanwhile, the single-class recognition accuracy of each gesture is also used to evaluate the difference between gesture categories.

The leave-one-out validation of eight subjects for ten gesture categories is shown in [Table III](#). The proposed method has the best classification performance for sub.6, with an accuracy of 98.89%, an average precision of 99.09%, an average recall, and an average F1-score of 99%. Similarly, sub.1, sub.2, sub.3, and sub.8 all achieved above 95% with regards to these four metrics. The result indicates that although personal habits lead to the ambiguity and variability of gestures, the proposed method has good generalization for gesture classification of different subjects. By averaging the metrics of each subject, the proposed method achieves a 94.58% accuracy, 94.66% average recall, 94.57% average F1 score, and 95.22% average precision.

TABLE III Leave-One-Out Cross-Validation Results Based on the Proposed Method

Sub.	ACC.(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)										
					a	b	c	d	e	f	g	h	i	j
Sub. 1	95.56	94.89	95.28	96.39	95	82.61	100	100	95	100	91.3	100	100	100
Sub. 2	96.11	95.71	95.88	96.54	95	100	100	94.12	100	91.3	100	100	100	85
Sub. 3	95	95.27	95.22	95.36	94.74	94.12	85.71	90	100	95.24	100	93.75	100	100
Sub. 4	88.33	88.55	88.36	89.01	87.5	75	66.67	87.5	86.67	94.44	100	100	92.31	100
Sub. 5	90	90.94	90.05	92.43	100	73.68	100	100	63.33	92.86	100	94.44	100	100
Sub. 6	98.89	99	99	99.09	100	100	100	100	100	100	100	100	100	90.91
Sub. 7	94.44	94.56	94.47	94.54	100	94.12	88.89	94.44	94.74	94.12	85	94.12	100	100
Sub. 8	98.33	98.33	98.33	98.38	100	95	93.33	100	100	100	100	95.45	100	100
Avg.	94.58	94.66	94.57	95.22	96.53	89.32	91.83	95.76	92.47	96.00	97.04	97.22	99.04	96.99

In addition, the recognition results of each gesture category show that the precisions of the six gestures are all above 95%, while only the precision of the draw-circle gesture is lower than 90%, and the precisions of pull-down and push-forward gestures are below 95% but above 90%. This is because the proposed method can achieve complementary advantages through the combination of RGB image data and RDM data.

Since camera and radar have different sensitivities to the physical properties, the single transmit and single receive antenna channel of the radar sensor is used to collect the radial range and velocity of the gestures, while the azimuth, elevation, and imaging information of the gestures are collected by the camera. The texture features in multimodal data are fully extracted by the loss function of Gram to further improve the representation of features. The network structure of the proposed method includes deformable convolution, which can enhance the generalization performance through a more flexible receptive field.

B. Verification of Six Environments

As gesture sensing scenarios expand, robustness has become extremely important for gesture recognition based in complex environments. We deploy six complex environments, as shown in [Fig.5](#). The open scene outside the laboratory leads to weak electromagnetic clutter, while a bio-radar for detecting human vital signs is placed inside the crowded laboratory to simulate measurement conditions with high environmental clutter. [Fig.5\(a\) and \(b\)](#) shows that energy dispersion and redundant targets occur in the RDM of the in-laboratory scene, which decrease the performance of the RDM data. Similarly, different lighting conditions are used to evaluate the adaptability of the proposed method to illumination. [Fig.5\(c\) and \(d\)](#) shows that the amount of information expressed by the RGB image is sharply decreased by the dark and bright light conditions.

Multimodal and single-modal data are collected by new subjects to implement gesture classification for the six environments. The network structure of HGR using single-modal data is the proposed method with a single stream. [Table IV](#) summarizes the results of HGR for the six environments using multimodal data of RGB images and RDM. Classification based on multimodal data has achieved 94.63% precision in the out-of-laboratory of normal light scenes. Under the in-laboratory of normal light scene, recognition performance, which drops slightly, is above 93%. Although electromagnetic

clutter in the in-laboratory scene impairs the expressiveness of the RDM data, the RGB image data can still provide sufficient reliable information for gesture classification. Regarding the dark conditions, depending on the intact RDM data, the HGR for the out-of-laboratory scene achieves 93.68% classification precision, while the precision of HGR for the in-laboratory scene is reduced but still achieves 89%. Similarly, with regards to brightly lit conditions, the classification results show that although the quality of the RGB image is significantly damaged, the precision of the proposed method still reaches more than 83%. This is because the proposed method fully exploits the complementary information of multimodal data to implement the HGR and prevents the influence of environmental factors on the classification performance.

TABLE IV Verification of Six Environments Using Multimodal Data

Env.	ACC.(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)										
Outdoor & Normal	94.03	94.5	94.48	94.63	92.06	98.11	90.62	89.83	95	96.77	100	90.77	96.61	96.49
Indoor & Normal	92.96	93.17	93.1	93.14	92.19	88.89	91.8	91.8	95.24	92.19	100	84.48	96.55	98.28
Outdoor & Dark	93.01	93.5	93.55	93.68	100	78.46	83.87	100	98.33	93.75	98.31	96.61	87.5	100
Indoor & Dark	89.37	89.67	88.81	89.15	98.33	72.86	84.75	96.43	96.55	85.48	98.25	82.14	98.25	98.25
Outdoor & Bright-light	89.29	89.33	89.44	89.9	85.29	70	90.74	100	92.06	88.89	100	91.67	80.36	100
Indoor & Bright-light	83.48	83.83	83.86	84	84.85	68.25	80.36	75	93.1	84.13	94.83	67.8	91.67	100

Confusion matrices of HGR for the six environments using multimodal data are shown in [Fig.6](#). The distribution of the confusion matrix indicates that the reduction in accuracy of most gestures caused by data damage is limited. The draw-circle and click gestures are influenced by the indoor and outdoor sceneries. This is because the weak spectral energy in the RDM of these gestures is easily disturbed by the indoor electromagnetic clutter. In addition, draw-circle and push-up gestures are also susceptible to illumination conditions. It seems that the recognition of these gestures relies on the azimuthal or elevational information, and the dark or bright light condition cannot provide reliable azimuth or elevation.

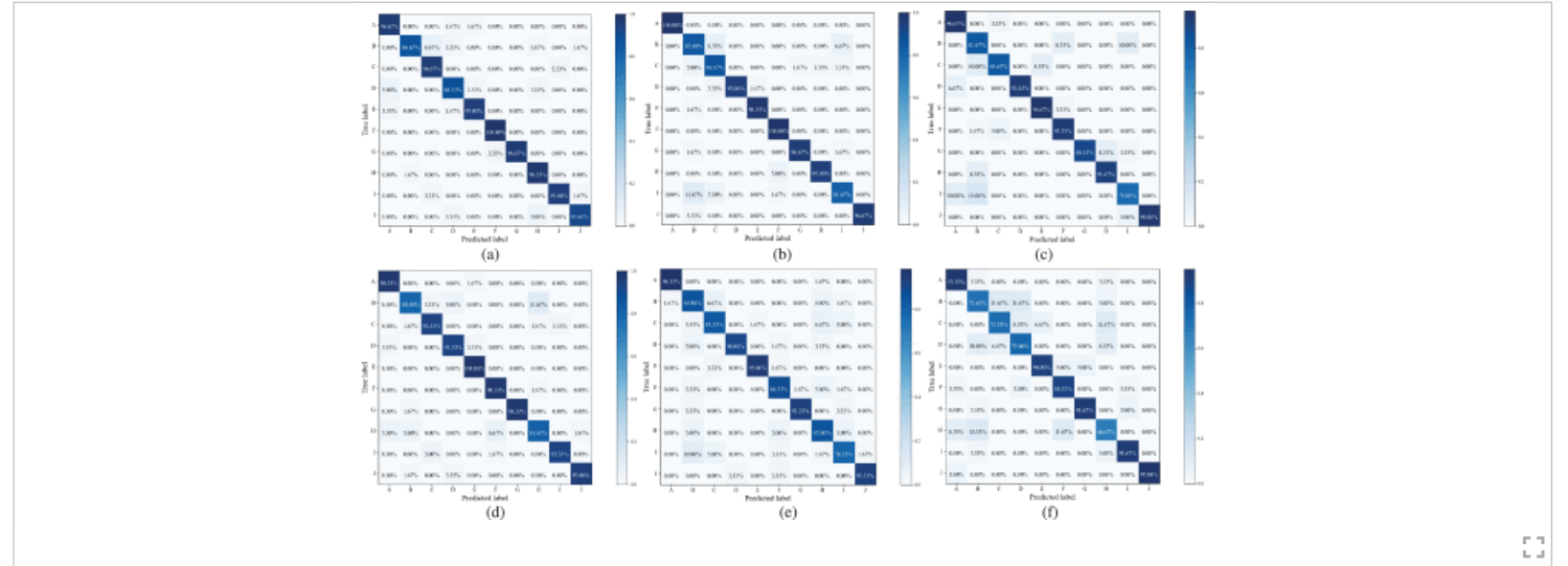


Fig. 6.

Confusion matrixes of HGR based on the six environments using the multimodal data. (a) Outdoor and normal. (b) Outdoor and dark. (c) Outdoor and bright light. (d) Indoor and normal. (e) Indoor and dark. (f) Indoor and bright light.

[Table V](#) summarizes the results of the HGR for the six environments using only RGB image data. Under normal illumination, the HGR based on RGB image data can achieve an accuracy of over 90% in both indoor and outdoor scenarios. However, under dark or bright light conditions, the gesture classification performance suffers a significant decline. The classification accuracy under bright light is only approximately 55%. Confusion matrices of the HGR for the six environments using RGB image data are shown in [Fig. 7](#). The distribution of the confusion matrix indicates that gestures distinguished by visual information, such as draw-circle, pull-down, pinch-hand, and pull-up, are difficult to recognize in complex lighting environments. Regarding the bright light condition, the accuracy distribution of each gesture category becomes irregular. Individual gestures still achieve good recognition performance, which may be caused by the fact that the larger shooting area of gestures makes the receptive field wider in the RGB image.

TABLE V Verification of Six Environments Using RGB Image Data

Env.	ACC.(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)												
					a	b	c	d	e	f	g	h	i	j		
Outdoor & Normal	91.75	92.17	92.12	92.42	89.23	89.23	89.29	85.31	93.44	100	100	96	91.23	90.32		
Indoor & Normal	90.85	91.23	91.27	91.84	87.1	87.88	96	95	93.44	100	100	93.75	100	84.06		
Outdoor & Dark	76.97	77	77.7	81.85	84.06	37.5	90.7	62.5	98.31	100	100	85.19	100	60.29		
Indoor & Dark	73.44	73.67	74.09	76.64	78.18	47.19	74.58	45.45	77.03	100	100	90.62	100	53.33		
Outdoor & Bright-light	58.01	57.83	54.43	58.44	57.14	100	58.62	27.27	25.93	61.7	91.8	10.81	68.29	82.81		
Indoor & Bright-light	55.78	55.33	49.68	51.87	47.86	100	37.07	0	0	75.34	100	0	93.44	65		



Fig. 7.

Confusion matrixes of HGR based on six environments using the RGB image data. (a) Outdoor and normal. (b) Outdoor and dark. (c) Outdoor and bright light. (d) Indoor and normal. (e) Indoor and dark. (f) Indoor and bright light.

[Table VI](#) summarizes the results of the HGR for the six environments using only RDM data. In the out-of-laboratory scene, although illumination has been transformed, a single RDM data can still robustly achieve the HGR, with an accuracy of more than 89%. However, the complex environmental clutter in the in-laboratory scene severely disturbs the energy distribution of the micro gesture features in the RDM data. A 15%–20% drop in the accuracy results in the gesture classification task not being completed properly. Confusion matrices of the HGR for the six environments using RDM

data are shown in [Fig. 8](#). The distribution of the confusion matrix indicates that the indoor electromagnetic clutter can significantly degrade the accuracy of gesture classification that relies on the radial motion information, such as pull-down, slide-left, click, pull-up, and wave, while the effect is much smaller for gestures with a high spectral energy, such as push-back and push-forward.

TABLE VI Verification of Six Environments Using RDM Data

Env.	ACC(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)										
Outdoor & Normal	91.33	91	91.01	91.35	91.94	96.43	85.45	81.68	100	92.98	100	91.94	78.69	95
Indoor & Normal	72.31	72.67	72.61	73.58	94.92	64.29	37.93	72.06	89.83	97.62	83.82	72.34	63.79	59.15
Outdoor & Dark	90.15	89.83	89.75	90.06	96.72	93.22	85.71	93.55	100	84.21	96.55	90.62	75.76	84.29
Indoor & Dark	71.73	72	71.82	72.85	82.19	84	45.28	76.27	93.1	97.56	84.29	56.36	50	59.42
Outdoor & Bright-light	89.05	89.35	89.2	89.34	88.71	93.33	75.47	98.25	92.06	87.5	83.33	94.74	83.33	96.67
Indoor & Bright-light	74.28	73.83	74.25	75.43	93.44	73.44	45.28	92	100	93.75	90.77	53.62	45.83	66.18

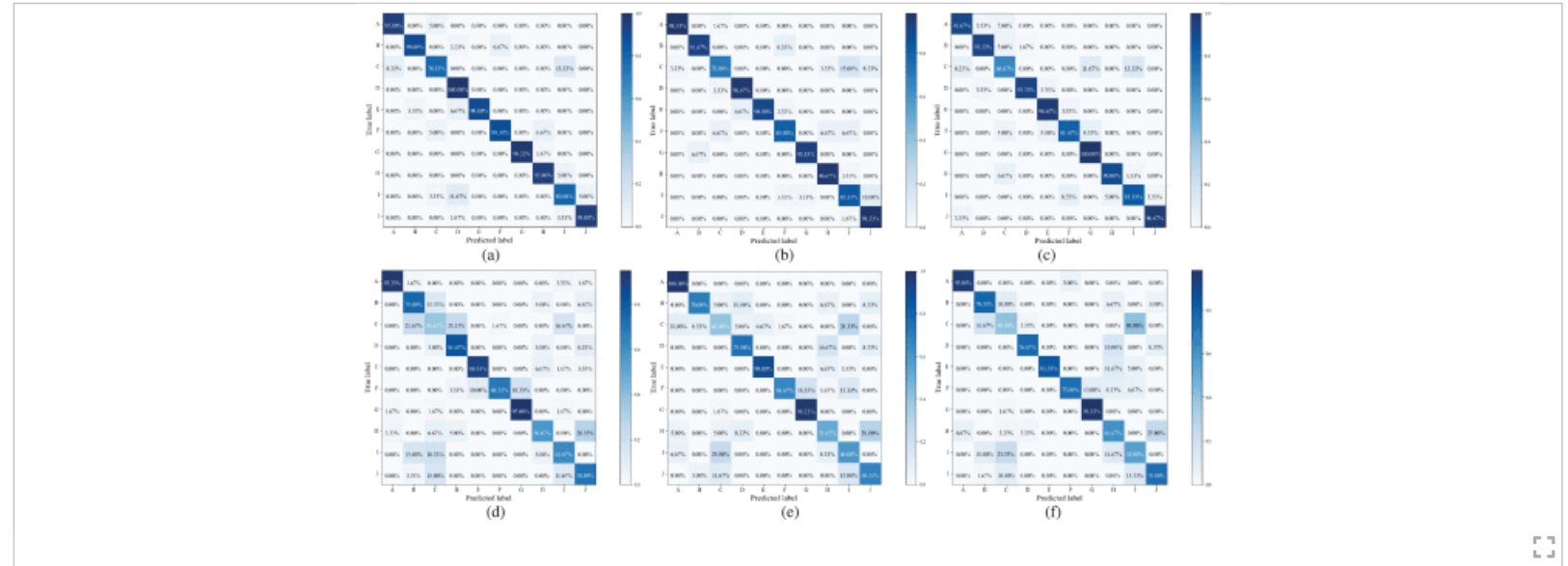


Fig. 8. Confusion matrixes of HGR based on six environments using the RDM data. (a) Outdoor and normal. (b) Outdoor and dark. (c) Outdoor and bright light. (d) Indoor and normal. (e) Indoor and dark. (f) Indoor and bright light.

C. Ablation Study

Gestures have different feature distributions in the multimodal data of FOV scenes with different ranges. In this work, the variation is addressed by embedding several network structures, such as deformable convolutions, Gram-based loss terms, and LSTM layers. Since different spatiotemporal networks are compared in [Section VI-D](#), the deformable convolution and the Gram-based loss are the main objects of this ablation study.

[Table VII](#) records the results of the ablation studies in different FOV scenes with normal light. When the subject collects gestures at 1 m, after training with the deformable convolution and the Gram-based loss function, the gesture classification accuracy can reach 94.58%. If the complementary connections between multimodal features have not been extracted by the Gram-based loss, the classification performance of gestures will inevitably drop, with an accuracy of 91.67%, an average precision of 91.19%, an average recall of 92.67%, and an average F1 score of 90.47%. If the deformable

convolution has been replaced by the ordinary convolution, the classification accuracy of the gesture will further drop to approximately 90%. When the subject collects gestures at 1.5 and 2 m, although the performance of the HGR gradually decreases with an increasing distance between the subject and the measurement platform, the proposed method with deformable convolution and Gram-based loss can still maintain the four classification metrics to be above 86%. When the subject collects gestures at 2.5 m, the four metrics of the network structure that remove the deformable convolution and Gram-based loss are reduced to approximately 75% and 80%, respectively. The network with both the structures can maintain the classification accuracy at approximately 85%.

TABLE VII Ablation Study in Different FOV Scenes

Different Range	W/ Deform.	W/ G-loss.	ACC.(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)
1 meter	×	✓	90.00	93.24	90.40	89.58
	✓	×	91.67	92.67	90.47	91.19
	✓	✓	94.58	94.66	94.57	95.22
1.5 meters	×	✓	90.00	90.95	90.26	92.35
	✓	×	88.33	89.21	88.24	88.17
	✓	✓	91.67	91.48	91.50	93.56
2 meters	×	✓	83.33	80.67	78.77	82.12
	✓	×	85.42	85.42	83.30	89.47
	✓	✓	88.33	87.86	86.94	92.98
2.5 meters	×	✓	75.00	78.17	73.52	78.75
	✓	×	80.00	78.05	76.44	81.60
	✓	✓	85.00	84.09	80.65	81.33

The above ablation studies demonstrate that deformable convolutions and Gram-based loss not only provide reliable gesture recognition but also enhance the generalization for gesture classification tasks in different FOV scenes.

D. Comparison of Different Networks

In our work, the proposed method relies on the two-branch shallow CNN and the LSTM layer to extract spatial and temporal features from multimodal data, respectively. To verify the feature extraction and classification capabilities of the proposed method, the HandNet presented by Wang et al. [65] to implement the radar-based HGR, the ResNe18t with deep learning structure [66], the RNN composed of the fully connected and LSTM layer [64], and the 3-D ResNet18 using time-dimensional convolution to extract temporal features [67] have been compared with the proposed method in Table VIII. Specifically, the shallow CNN is replaced by the HandNet, ResNet18, and fully connected layer, and the outputs of the two-branch 3-D ResNet18 are concatenated for the HGR.

TABLE VIII Comparison of Different Networks

Networks	ACC(%)	Avg. Recall(%)	Avg. F1-score(%)	Avg. Prec.(%)
HandNet [65] (Two-branch+LSTM)	83.33	82.43	79.51	79.04
ResNet [66] (Two-branch+LSTM)	88.15	88.47	87.62	90.15
3D ResNet [67] (Two-branch)	88.89	89.91	86.95	93.46
RNN [64] (FC+LSTM)	84.17	84.26	83.57	87.64
Proposed Method	94.44	94.72	94.73	95.36

In Table VIII, the proposed method achieves the best recognition results among the four metrics. ResNet18-LSTM and 3-D ResNet18 using deep convolutional networks achieve similar classification accuracies, which are close to 90%. However, HandNet only achieves a classification accuracy of

83.33%, and the F1-score is lower than 80%. The RNN method, which achieves a classification accuracy of 84.17%, also cannot compete with convolutional neural networks.

In addition, [Figs. 9](#) and [10](#) show the training process of these methods. The red line in the figure represents the training process of the proposed method. As shown in [Fig. 9](#), although the loss value is higher in the first ten epochs due to the Gram-based loss, the proposed method also facilitates the extraction of heterogeneous information from multimodal data and converges the loss value below 0.1 at epoch 30. In contrast, the loss values of HandNet and RNN remain above 0.5 at epoch 50. [Fig. 10](#) indicates that the four metrics of the proposed method have reached more than 90% at epoch 25 and finally converged to a precision of 95.36%, which is significantly better than other networks.

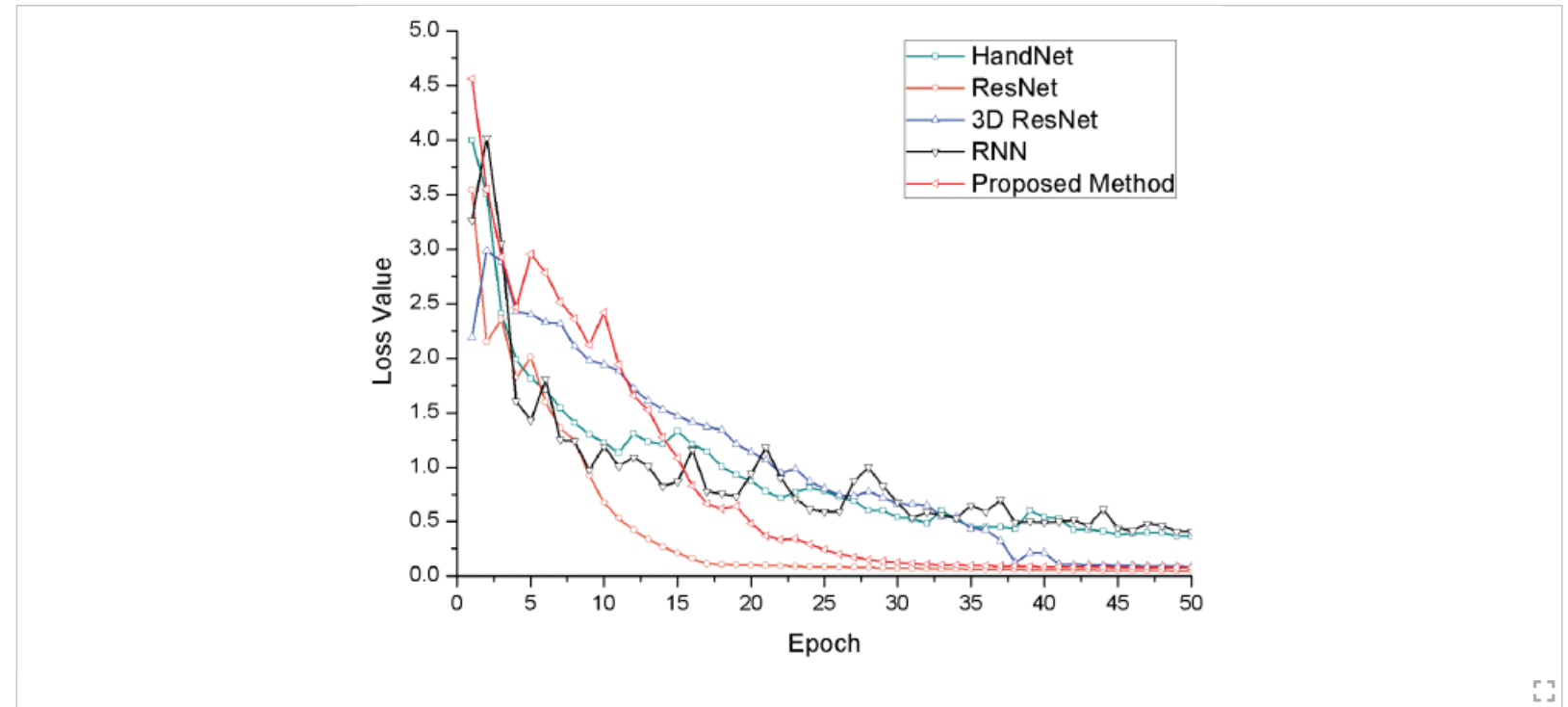


Fig. 9. Different network-based training processes of loss value. The red line in the figure represents the training process of the proposed method.

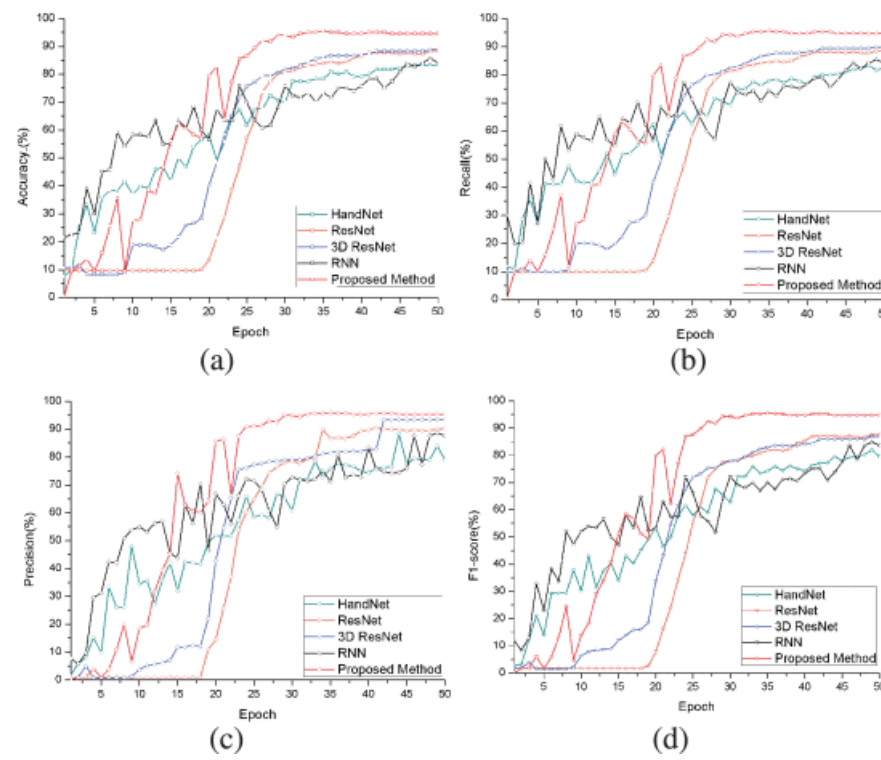


Fig. 10.

Different network-based training processes of accuracy, average recall, average F1-score, and average precision. The red line in the figure represents the training process of the proposed method. (a) Training process of accuracy. (b) Training process of recall. (c) Training process of precision. (d) Training process of F1-score.

E. Discussion

In this work, we try to address the issue that a single sensor cannot reliably achieve HGR in complex environments and different FOV scenarios. A multimodal measurement platform composed of radar and camera is used to collect gesture data. The well-designed comparison experiments show the recognition performance of ten gestures in different environments. The proposed multimodal method achieves relatively stable performance for the six environments.

Regarding outdoor environments, radar data can always provide support for gesture classification regardless of lighting changes. Regarding indoor environments, radar performance is degraded by background noise. The weak spectrum in the RDM of individual gestures, such as draw-circle (b) and click (h), is easily disturbed by indoor background noise. Some gestures that rely on radial features, such as push-forward (a), push-backward (e), and waving (j), are also affected to varying degrees. In the dark environment, some gestures, such as slide-left (f), slide-right (g), pull-up (I), and pull-down (c), have a certain deviation in the classification results. In the bright light environment, gestures with a good classification performance also experience a decrease in accuracy.

Gesture classification results from a single sensor can better reflect the dependence of gesture data from different sensors in complex environments. From these experiments, it is evident that multimodal data have a high advantage over single sensor data. Collecting data in a variety of environments and scenarios makes our experimental results rich and convincing. In the future, we will collect more subjects to build a complete multimodal gesture dataset.

Our work does not require spatial synchronization since the classified objects are dynamic gestures. In radar data, as the gesture changes, the size and position of the energy scattering center of the RDM from different frames will change. Regarding vision data, images from different frames can show displacement variations in gesture key-points. The proposed time synchronization preprocessing method is used to extract and segment the start and end times of each gesture piece from the multimodal data with different start time stamps. Each slice is labeled according to the gesture items performed by the subject, thus obtaining camera and radar gesture data with the same starting time stamp.

We deploy the measurement platform with separate radar and camera sensors. First, by adjusting the sampling frequencies of the radar and camera, the scales of the data generated by the two sensors agree within 1 s. Then, the camera and radar are turned on separately for object detection. After the gesture acquisition is complete, we turn off both the sensors and derive gesture data with different physical properties. It is worth noting that the input data sizes of the existing studies are usually between 16×16 and 96×96 . Therefore, considering the training cost of the model, we chose a data size of 32×32 . In future work, we will further investigate how to maintain classification performance in memory-constrained edge cases.

SECTION VII.

Conclusion

In this article, we propose a novel multimodal dynamic HGR method based on a two-branch fusion deformable network with Gram matching to address the unreliable HGR caused by complex environments and differences in personal habits. The experimental results indicate that the proposed method effectively improves the adaptability of the classifier to complex environments and exhibits satisfactory robustness to multiple subjects. Ablation analysis shows that deformable convolution and Gram loss not only provide reliable HGR but also enhance the generalization ability of classification tasks in different FOV scenarios. Overall, the experiments show that the proposed radar–vision fusion-based dynamic gesture recognition method is a robust solution that is applicable to complex environments with strong generalizations.

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